

Sumit Raj

CONTACT

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Portfolio: sumit966-github-io.vercel.app

SKILLS

Languages: Java, Python, JavaScript, C++, SQL
Backend: REST APIs, FastAPI, Node.js, Object-Oriented Programming, Microservices (basic)
Databases: MySQL, PostgreSQL, MongoDB, SQLite
DevOps and Cloud: Docker, GitHub Actions CI/CD, Git, Linux, GCP
Fundamentals: Data Structures and Algorithms, SDLC

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur 2024 – May 2026
M.Tech, Applied AI and Machine Learning
Dr. D.Y. Patil Institute of Technology, Pune 2019 – 2023
B.E., Computer Science CGPA: 7.89/10
Certifications: Google Cloud: Cloud Fundamentals and Cloud Security; HackerRank: 5-Star Java, 4-Star C++, 3-Star Algorithms; 300+ coding problems solved

EXPERIENCE

Software Engineer Intern, Salesforce (Remote) Feb 2022 – May 2022

- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%.
- Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%.
- Designed secure data-management modules using object-oriented programming and secure development practices.

PROJECTS

RAG-Based Clinical Question Answering System 2025
Python, FastAPI, LangChain, ChromaDB, Docker, GitHub Actions

- Built REST APIs with FastAPI that serve LLM answers over clinical documents, reaching 92% answer relevance on a 100-question benchmark.
- Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%.
- Containerized with Docker and set up GitHub Actions CI/CD, keeping inference latency under 500 ms.

CI/CD Pipeline with Docker and GCP

GitHub Actions, Docker, GCP Compute Engine

- Containerized a Python web application with Docker and deployed it on GCP Compute Engine through a GitHub Actions pipeline, reducing manual deployment time by 70%.

Military Vehicle Detection and Face Authentication Security System

Academic Project

Python, YOLOv8, OpenCV, Haar Cascade, Docker, GCP Cloud Run

- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy.
- Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.
- Containerized the YOLOv8 system with Docker and deployed it on GCP Cloud Run for serverless inference, with Cloud Logging for monitoring and alerts.

Brain Tumor Detection from MRI Scans

2024

Python, Deep Learning

- Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.

Mobile Botnet Detection System

2024

Python, Scikit-learn

- Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.